## Class 25: Fermi Energy, Fermi Surface, Fermi Temperature

In the last class we noted that density of occupied states $N(E)$= the density of allowed states $D(E)$, times the probability of occupancy of the states $f(E)$

The density of occupied states therefore gives a more complete picture of the electrons in a solid.

We also noted that, at 0 K, when the nearly free electrons in the solid are used to fill up the available energy levels, in accordance with the rules applicable to Fermions, the energy levels are filled up starting from the lowest energy level upwards. The energy level that is filled up by the last available nearly free electron, is referred to as the Fermi Energy $E_f$.

We found that the energy of the electron relates to its wave vector through the relationship:

$$E = \frac{\hbar^2 k^2}{2m}$$

Where $k$ is the wave vector, and in a one dimensional case is the same as $k_x$. The wave vector $k$ has the dimension of $L^{-1}$.

Since the system can be looked at from the perspective of different variables, we can obtain plots of different topologies that convey similar or the same information. Constant energy will appear as a flat surface in some plots that have energy on one of the axis, but with other variables, a spherical surface may correspond to constant energy.

We can make a plot using the components of the wave vector $k$. As indicated earlier, in a one dimensional case $k = k_x$. In three dimensions, the magnitude of the wave vector will be such that $k^2 = {k_x}^2 + {k_y}^2 + {k_z}^2$.

Since:

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left({k_x}^2 + {k_y}^2 + {k_z}^2\right)$$

Therefore, similar to the quantum number space that we discussed in the previous class, in $k$ space, points corresponding to constant energy will lie on the surface of a sphere.

As we fill up the available states with the nearly free electrons, the electrons having the same energy will be represented by a sphere. Corresponding to electrons of higher energies, there will spheres of larger radii. Corresponding to the electrons with the highest energy in the system, which is the Fermi energy $E_f$ at 0 Kelvin, the sphere that will be generated is called the Fermi surface.

Figure 25.1 below shows a schematic of the Fermi surface and the corresponding plot of density of occupied states. The Fermi energy shows up as a straight line in the density of states plot, since energy is on the x-axis. In the plot in *k* space, the outermost sphere corresponds to the Fermi energy.

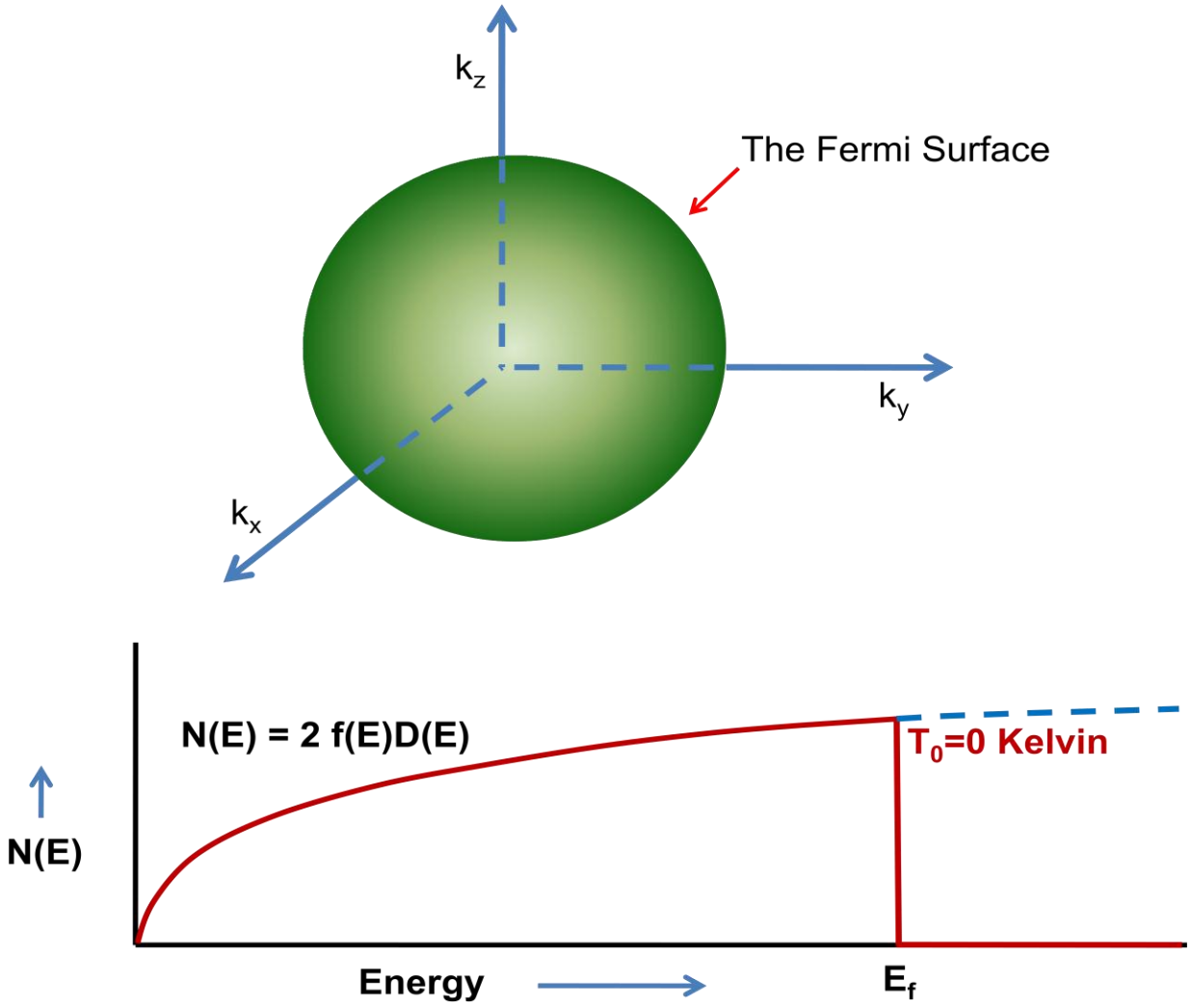



**Figure 25.1:** Plot of allowed states in *k* space, and the corresponding density of states plot. Fermi energy is represented by a sphere in *k* space, called the Fermi surface, and as a straight line in the density of states plot.

Based on the system we choose, the $E_f$ can differ. Even within the same system, the $E_f$ can change if there is a change of phase that affects the number of free electrons per unit volume.

We have therefore identified two concepts thus far: The energy at which we run out of free electrons while filling up the energy levels at 0 Kelvin, which is the Fermi energy, and its

corresponding surface in $k$ space, the Fermi surface. At the moment these are only definitions. The significance and use of these concepts we will see a little latter.

While examining the translational kinetic energy of classical particles, we found that:

$$\frac{1}{2}mv^2 = \frac{3}{2}k_bT$$

While we recognize that we are not dealing with classical particles, we can still utilize the idea that if the energy of the particle is considered to be thermal energy, then it is related to the temperature $T$ consistent with that thermal energy through the expression $k_bT$. Therefore, corresponding to the electrons at the Fermi energy $E_f$, a temperature can be defined, which is called the Fermi temperature $T_f$, such that:

$$E_f = k_bT_f$$

We have so far identified three quantities:

1) Fermi energy $E_f$
2) Fermi surface
3) Fermi Temperature $T_f$

Let us now examine their significance.

**Fermi Energy:**

To understand its significance of Fermi energy, let us compare it with temperature, pressure, and chemical potential. In general, we can think of temperature as a measure of the tendency to push heat out of a given location. Therefore, higher the temperature, greater is the tendency to push heat out of that given location. When a hot body comes in contact with a cold body, heat is pushed from the hotter body to the colder body. Therefore heat flows from the hot body to the cold body. Heat continues to move from the hot body to the cold body till the temperatures of the two bodies equalize, at which point the tendency to push heat out of either of the bodies is the same, and thermal equilibrium is attained.

Similarly, pressure represents the tendency to push material away from a location in the system. When two containers differing in pressure are connected to each other, material moves from the container at higher pressure to the container at lower pressure, till the tendency to push material out of either container is the same, and hence equilibrium is attained.

Chemical potential is associated with each specific species in a phase, and represents the tendency of the specific species to be pushed out of a specific phase. When two phases come in contact with each other, any given species will move from the phase where it has a higher chemical potential to the phase where it has the lower chemical potential. Chemical equilibrium

is attained when the chemical potential of all of the individual species, is the same in all of the phases in contact.

Electrons can be thought of as one of the species that go to make up the material. When two materials come in contact with each other, if electrons in one of the materials have higher energy, they will tend to flow into the material where they have lower energy. Since electrons in a material fill up a range of energy levels, which energy should be used for the comparison? Since all of the energy levels up to the Fermi energy are filled in the ground state of the material, the most appropriate energy to use for comparison between electrons in two materials is the Fermi energy itself. It is similar to comparing the highest level of water in two containers to determine which direction the water will flow in when the two containers are connected. In this sense, the Fermi energy of electrons is analogous to the chemical potential μ, of the electrons.

$$E_f \approx \mu$$

The equality is quite reasonable at low temperatures, but becomes less accurate at very high temperatures since $E_f$ represents the highest occupied energy level for metals only at 0 Kelvin.

The identification of $E_f$ with the chemical potential of electrons, is very useful since it helps us predict what will happen when two dissimilar materials are brought in contact. In the electronics industry junctions between different semiconductors are common, and knowledge of the $E_f$ of the two materials assists in determining the direction of natural flow of electrons due to the formation of the junction. Therefore, knowledge of $E_f$, has significant practical applications.

**Fermi surface:**

Fermi surface is the surface in $k$ space that corresponds to the Fermi energy. In quantum mechanics we accept the wave particle duality and associate particles with a waves consistent with their momentum. In $k$ space, we are plotting wave vectors $k = \frac{2\pi}{\lambda}$, where $\lambda$ is the wavelength corresponding to electrons with specific energy. Therefore, the Fermi surface represents wave vectors of electrons having the Fermi energy $E_f$. To understand the significance of this, let us briefly look at the interaction of radiation with matter.

When radiation in the form of X-rays or an electron beam, interacts with matter, we see diffraction effects. Figure 25.2 below shows a schematic of a typical experiment to record diffraction patterns.

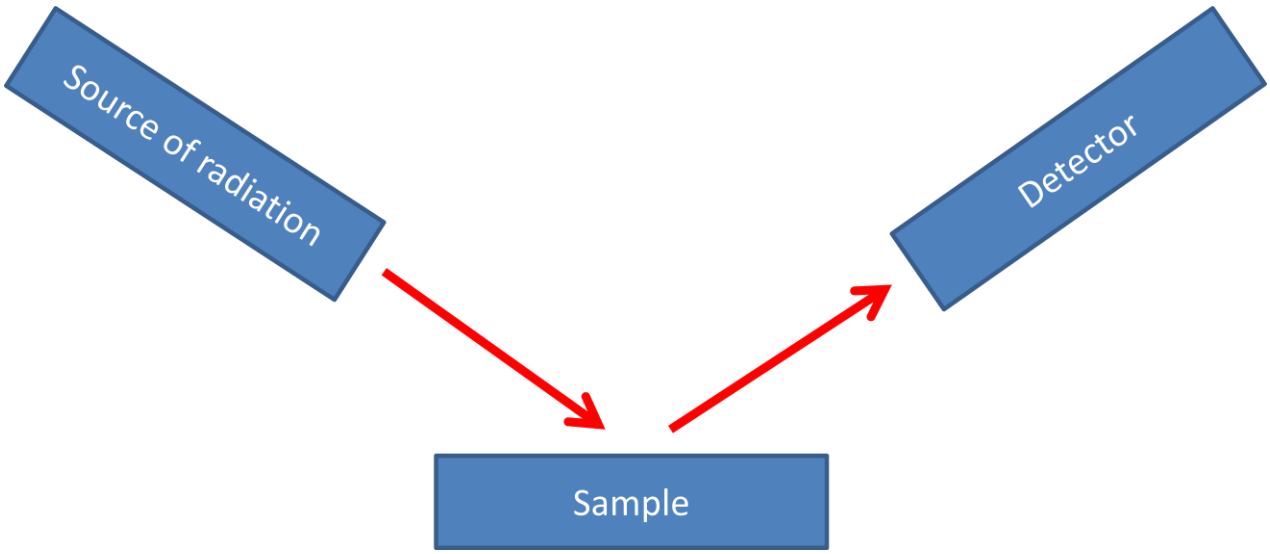



**Figure 25.2:** A schematic showing how radiation can be made to interact with matter to produce and record diffraction effects in a laboratory setting.

To enable recording diffraction effects, in general we need:

1) A source of radiation
2) Sample with a periodic structure
3) Wavelength of the radiation to be in a range that enables diffraction effects to be recorded (which is of the order of the spacing of the periodic structure, or less)

However, the important aspect to note is that even though the schematic shows the source of the radiation to be independent of the sample, this is only an experimental convenience. Waves of electrons already within the sample, can interact with the periodic structure of the sample and display diffraction effects. Unlike a separate electron beam source in an electron microscope, this is the equivalent of an electron beam within the sample.

The Fermi surface represents the wave vectors of the electrons corresponding to the Fermi energy. $k$ space plots this information in inverse length dimensions. We can plot the periodic structure of the material also using inverse length notation, on the same plot. This combined plot can help us identify diffraction conditions for the nearly free electrons within the material due to interaction with the periodic structure of the same material. The plot of the periodic structure of the material in inverse length dimensions is referred to as a plot in 'Reciprocal space' – which will be examined in greater detail in a subsequent class, and is a very important tool in understanding the behavior of electrons in solids. For the moment we will simply state in summary that the diffraction of nearly free electrons results in the formation of permitted energy levels and forbidden energy levels, or bands for the free electrons – a more detailed account of this interaction will also appear in a subsequent class.

**Fermi temperature:**

The Fermi temperature $T_f$, is defined as the temperature corresponding to the Fermi energy such that:

$$E_f = k_b T_f$$

With the Fermi energy of the order of a few eV, the Fermi temperature works out to be of the order of several thousand Kelvin (approximately 10,000 K). How do we reconcile the fact that the material is at room temperature or lower, or even at 0 Kelvin, while the Fermi temperature is 10,000 K? One way to look at the situation is as follows: When we measure the temperature of a material, we do not typically measure the temperature of a single atom or electron. What we measure is the average temperature of the material. There is invariably going to be a distribution of energy within the material. In this distribution, an extremely small thermal mass, consisting of a very small fraction of the nearly free electrons (which is itself a very small fraction of the total electrons in the system), is at the Fermi energy, and the temperature corresponding to that energy is the relatively high Fermi temperature. Therefore the 'high' Fermi temperature is not inconsistent with the 'low' temperature or the solid as a whole.

In the next class we will make use of the Fermi temperature to understand how corrections can be made to the estimate of the $C_V^e$.